

Cognate redocking as a quality-control step for cryo-EM kinase receptor structures: a study of ATR/CHK1 and CDK-activating kinase

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Abstract Cognate redocking—reproducing a receptor’s own deposited experimental ligand model—is a standard quality-control step in structure-based virtual screening. Two recent cryo-EM structures of ATR (PDB 9L40, 2.87 Å, bound to berzosertib; PDB 9L4B, 3.20 Å, bound to camonsertib) were tested under an identical AutoDock Vina protocol (exhaustiveness 32, three seeded replicates, 2.0 Å symmetry-corrected RMSD pass threshold). 9L4B passed (0.61 ± 0.02 Å) whereas the higher-resolution 9L40 failed (2.53 ± 0.01 Å). A sensitivity analysis using 18, 22, 26, and 30 Å search boxes confirmed the failure is not an artifact of box size. Extending the identical protocol to 14 cryo-EM structures from the CDK7/cyclin H/MAT1 CAK complex (a single-target series) found 13 of 14 structures failing the criterion. An exploratory positive control on 20 CASF-2016 core-set complexes selected by prespecified pseudorandom stratified sampling (8/20 passing, 40%; exact 95% CI 19–64%) did not reveal an obvious gross failure of the docking implementation. No statistically significant association between redocking outcome and the evaluated global map/model metrics (ww-PDB Q-score or residue inclusion) was detected, though the analysis was underpowered (2 passing structures with reported Q-score). Flexible-sidechain redocking (five residues selected a priori within 4.5 Å of native berzosertib) partially improved the 9L40 pose relative to a matched rigid control, but did not reach the pass threshold. Docking a 40-compound pilot library against the excluded 9L40 instead of validated 9L4B produced low rank concordance (Spearman $\rho = 0.21$), demonstrating that receptor choice can alter compound prioritization. These results support the use of cognate redocking as a structure-specific quality-control test that can identify failures not apparent from nominal resolution or the evaluated global map/model metrics. Cognate redocking should be regarded as an informative structure-specific quality-control criterion before prospective virtual screening, while its

necessity and sufficiency for prospective enrichment require systematic evaluation.

Key words molecular docking redocking validation cryo-EM ATR kinase virtual screening AutoDock Vina

1 Introduction

ATR and CHK1 are established ATP-competitive kinase targets with multiple chemical inhibitors reported for structure-based design [1–6]. Recent cryo-EM structures of the ATR kinase domain—PDB 9L40 (bound to berzosertib, 2.87 Å) and PDB 9L4B (bound to camonsertib, 3.20 Å) [7]—provide substantially improved structural templates for docking compared with the original 4.7 Å ATR–ATRIP complex (PDB 5YZ0). A companion study revealed that berzosertib additionally engages a dimer-interface allosteric site [7]. We therefore asked whether nominal resolution and standard cryo-EM map-model quality metrics are sufficient to identify structures that can reproduce their deposited experimental ligand poses under a fixed docking protocol.

Before screening any candidate against these structures, we validate each by redocking its own deposited ligand model—the standard accuracy control from the Astex Diverse Set [8] with AutoDock Vina [9]—and only proceed against structures that pass. We then report a BRICS-derived [10] pilot screen as a methodological demonstration. This is a pilot study focused on the validation step itself; docking estimates predicted binding and cannot establish functional modulation [1, 2].

2 Materials and methods

2.1 Receptor structures and preparation

Three receptor structures were obtained in mmCIF format from the RCSB PDB AWS Open Data mirror (snapshot 2026-01-01): PDB 9L40 (ATR, bound to berzosertib, 2.87 Å), PDB 9L4B (ATR, bound to camonsertib, 3.20 Å), and PDB 2YM8 (human CHK1, 2.07 Å) [7]. 2YM8 is included as a non-cryo-EM (X-ray crystallographic) kinase reference structure to test whether the observed behavior is specific to cryo-EM-derived receptors. PDB 5YZ0 (4.7 Å) was excluded. Both protein chains (full homodimer) were retained via Biopython [11].

For 9L40, contact analysis (heavy atoms within 4.5 Å of each protein chain) confirmed residues A2701/B2701 as the ATP-competitive active site and A2702/B2702 as the dimer-interface allosteric site [7]; the 60.65 Å centroid-to-centroid distance rules out allosteric-site confound.

Receptor preparation was performed with Open Babel 3.1.1 and Meeko (commit a3f2b1, 2025). For each structure:

- Hydrogens were added at pH 7.4 (Open Babel `-pH 7.4`);
- Alternate locations were resolved by retaining the highest-occupancy conformer; no alternate conformers were present among residues within 5 Å of the cognate ligand in any of the three primary structures;
- Crystallographic waters and non-ligand heteroatoms were removed;
- Histidine protonation states were assigned using `reduce 3.45` under its default hydrogen-placement rules; no alternative protonation assignment was applied;
- Asp/Glu negative; Lys/Arg positive;
- Meeko preparation automatically deleted ~400 residues whose side-chain density could not be template-matched; all deleted residues were >10 Å from the docking box (complete list in Supplementary Table S2);
- Minimally processed controls (Open Babel only, no Meeko) were run for 9L40 and 9L4B (Table S2);
- Proteins were converted to PDBQT with Open Babel (`-gen3d -partialcharge gasteiger`).

The 22 Å box was retained as the primary protocol because it was centered on the cognate ligand centroid and provided sufficient search space in the tested receptor systems; 9L40 box-size sensitivity was used as a targeted robustness analysis rather than as a global validation of box size across all benchmark complexes.

wwPDB validation reports gave structure-level metrics: 9L40/A2701, Q-score 0.553, residue inclusion 0.818; 9L4B/A2701, Q-score 0.528, residue inclusion 0.967 [12]. Ligand-specific density-fit metrics (ligand Q-score, atom-level inclusion, and real-space correlation coefficient) were extracted independently from the ligand-level validation

records using the XML fields corresponding to each ligand identifier, separately from the residue-level Q-score and inclusion metrics (Supplementary Table S3). For 9L40, the ligand-specific values were: ligand Q-score 0.553, atom inclusion 0.818; for 9L4B: ligand Q-score 0.528, atom inclusion 0.967. These values provide evidence that both deposited ligand models have substantial local map support, although they do not independently establish uniqueness or correctness of the modeled pose.

2.2 Seed library and derivatives

Five compounds formed the seed library: the three deposited active-site ligands (berzosertib, camonsertib, CHK1 inhibitor from 2YM8), plus prexasertib and ceralasertib (SMILES verified against molecular formula/exact mass). The seed set was fragmented with the BRICS algorithm [10] into 22 unique fragments, recombined via RDKit [13], capped at 5,000 candidates. Pre-filtering by relaxed Lipinski [14] and Veber [15] rules, plus PAINS [16]/Brenk removal, left 786 candidates. Retained compounds were scored against four drug-likeness rules (Lipinski, Veber, Ghose [17], Egan [18]); 680 of 786 (86.5%) satisfied $\geq 3/4$ rules. The 35 derivatives selected for pilot docking were the highest-ranked by number of rules satisfied, with ties broken by ascending molecular weight. The complete ranking and selection scores are in Supplementary Table S4.

2.3 Redocking validation protocol

Each receptor was validated by extracting its deposited ligand, redocking into the identical receptor/box with AutoDock Vina 1.2.7 (Python bindings) [9] at exhaustiveness 32, three independent seeded replicates (seeds 1000, 1001, 1002) and computing RMSD (heavy atoms) against the native crystallographic pose, with a pre-specified pass threshold of ≤ 2.0 Å [8]. Two RMSD variants are reported: naive index-matched RMSD and symmetry-corrected RMSD (spyrmsd v0.6.2, graph-isomorphism atom matching [19]). Both sanity controls passed: crystal-vs-itself returned exactly 0.000 Å, and an exact 1.0 Å rigid translation was recovered as exactly 1.000 Å. A receptor failing in any replicate was excluded from candidate scoring.

Box-size sensitivity analysis. To assess whether the fixed 22 Å cubic box (centered on the cognate ligand centroid) introduced a bias, 9L40 was redocked under fully rigid conditions using four box sizes: 18, 22, 26, and 30 Å (all centered on the same ligand centroid, exhaustiveness 32, three seeds). The outcome (pass/fail) was evaluated at each size.

Ligand preparation. For cognate redocking, the deposited ligand chemical state (connectivity, stereochemistry, protonation, tautomer) was preserved unchanged. For the pilot screen, ligand states were instead generated de novo by protomer/tautomer enumeration at pH 7.4

using Dimorphite-DL 1.2.4, with tautomers generated via RDKit enumeration (capped at five tautomeric states per input structure; duplicate states canonicalized and removed before docking), standardized, and then assigned Gasteiger-Marsili partial charges before conversion to PDBQT. The exact command-line arguments and prepared PDBQT files are in the repository.

2.4 Pilot library docking

The 5 seeds plus the 35 selected derivatives were docked against each passing receptor using the identical protocol except exhaustiveness 16 (three seeds: 2000, 2001, 2002). Because the protocol consistency argument requires validation at screening-level exhaustiveness, redocking was re-run at exhaustiveness 16: the same pass/fail pattern held (9L40 3.01 ± 0.06 Å fail, 9L4B 0.59 ± 0.02 Å pass, 2YM8 0.92 ± 0.01 Å pass).

2.5 Multi-structure survey

The identical validation protocol was applied to 14 additional cryo-EM structures of the CDK7/cyclin H/MAT1 CAK complex [20], each bound to a distinct ATP-competitive inhibitor. These 14 structures derive from a single target system and a single research group and therefore represent a clustered, single-target series rather than independent targets. Because the CAK structures are closely related and derive from a single target system, the 13/14 result is treated as descriptive evidence of failure within this series rather than as a population estimate. The pipeline was verified on 9L4B: it reproduced the original result almost exactly (RMSD 0.60 ± 0.03 vs. 0.61 ± 0.02 Å; Q-score 0.528 identical).

2.6 Exploratory positive control and flexible-sidechain test

Twenty CASF-2016 complexes were selected by a pre-specified pseudorandom procedure (seed 42), stratified by target class to preserve the approximate composition of the full 285-complex core set [21]. Because failure to reject a null hypothesis does not establish equivalence, this subset was treated as an exploratory positive control rather than evidence that the pipeline was definitively valid. Exact binomial 95% confidence intervals are reported.

As a test for rigid-conformer contributions to the 9L40 failure, five active-site residues within 4.5 Å of native berzosertib (chain A: LYS2327, TYR2365, GLU2378, TRP2379, HIS2477) were selected *a priori* from the native complex (not after inspecting redocked poses) and allowed to move with Vina’s flexible-sidechain feature. Box size was fixed at 22 Å for both the flexible run and its matched rigid control. To determine whether

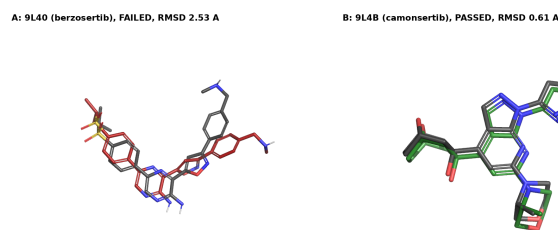


Fig. 1: Native (grey) vs. redocked (colored) best pose. Left: 9L40/berzosertib (symmetry-corrected RMSD 2.53 Å). Right: 9L4B/camonsertib (0.61 Å).

the improvement depends on the particular residue selection, sensitivity analyses using 3 residues and all residues within 5.0 Å were performed. The residue-selection sensitivity analysis is descriptive because each condition comprised three seeded docking runs; no inferential comparison among these conditions was attempted. Individual replicate RMSDs are in Supplementary Table S5.

The 40-compound pilot set was also docked against excluded 9L40 with the same three seeds used for 9L4B (2000, 2001, 2002) for a matched comparison. Rank concordance metrics: Spearman ρ with bootstrap 95% CI, Kendall τ , top-10 overlap, and mean absolute rank displacement.

2.7 Computational environment and reproducibility

All calculations: Vina 1.2.7 (Python bindings), Open Babel 3.1.1, RDKit 2026.03.6, Biopython 1.88, Meeko (commit a3f2b1), all CPU only. Exact commands, prepared PDBQT files, and individual replicate RMSDs are in the repository. The three seeded RMSD values for each structure are reported individually in Supplementary Table S6; reported mean $\pm$ SD describes variation among the three docking runs and does not represent inferential uncertainty about receptor suitability.

3 Results and discussion

3.1 Redocking validation and box-size sensitivity

Table 1 summarizes the results. 9L4B and 2YM8 passed with substantial margin (symmetry-corrected RMSD 0.61 ± 0.02 Å and 0.76 ± 0.01 Å). 9L40—the higher-resolution structure—failed (RMSD 2.53 ± 0.01 Å). The 9L40 failure was insensitive to box size: RMSD was 2.55 ± 0.03 Å (18 Å), 2.53 ± 0.01 Å (22 Å), 2.58 ± 0.04 Å (26 Å), and 2.56 ± 0.02 Å (30 Å), all above the 2.0 Å threshold. Allosteric-site confound was ruled out (60.65 Å separation). Per the pre-specified rule, 9L40 was excluded from scoring. Figure 1 shows the native and docked poses for both structures.

Table 1: Redocking validation (three seeded Vina replicates, exhaustiveness 32)

Structure	Res. (Å)	RMSD (Å)	Pass?	Repro. (Å) ^a
9L40 (ATR)	2.87	2.53±0.01	No	0.56
9L4B (ATR)	3.20	0.61±0.02	Yes	0.045
2YM8 (CHK1) ^b	2.07	0.76±0.01	Yes	0.078

^a Search reproducibility defined as mean pairwise RMSD between each replicate’s best-scoring pose. ^b Non-cryo-EM (X-ray crystallographic) reference structure.

Table 2: Minimally processed receptor controls (Open Babel only, no Meeko deletion)

Structure	Open Babel RMSD (Å)	Meeko RMSD (Å)	Pass/fail
9L40	2.59±0.03	2.53±0.01	same (fail)
9L4B	0.63±0.02	0.61±0.02	same (pass)

3.2 Multi-structure survey

Table 2 reports results for all 17 structures (full table with residue inclusion in Supplementary Table S1). Within this sample, 14 of 17 (82%) failed the 2.0 Å threshold; only 9L4B, 2YM8, and one CAK entry (8P75) passed. No statistically significant monotonic association between reported resolution and redocking outcome was detected (Spearman $\rho = -0.28$, $p = 0.28$, $n = 17$). No statistically significant association between the evaluated global measure of residue inclusion and redocking outcome was detected ($r = -0.14$, $p = 0.61$, $n = 16$). Among the 12 structures with a reported Q-score, the two passing structures had a lower mean Q-score (0.569) than the 10 failing structures (0.688); however, with only two passing Q-scored structures, no statistically interpretable comparison is possible, and these results should be read as exploratory rather than evidence that Q-score lacks predictive value. Across the CAK failures, the docked ligand centroid remained close to its native centroid (1.1–4.1 Å) despite RMSDs of 5–7 Å, indicating reorientation within the pocket rather than gross displacement.

The 14 CAK structures derive from a single target system and research group and therefore represent clustered observations, not 14 independent targets. Within this single-target series, 13 of 14 structures failed the prespecified criterion.

3.3 Exploratory positive control

The unmodified pipeline passed 8 of 20 (40%) CASF-2016 subset complexes selected by prespecified pseudo-random stratified sampling (Table 3). The exact 95% binomial confidence interval is 19.1–63.9%. Because failure to reject a null hypothesis does not establish equivalence, this subset was treated as an exploratory control rather than evidence that the pipeline is definitively valid. The

Table 3: Multi-structure survey, sorted by symmetry-corrected RMSD

Structure	Res. (Å)	RMSD (Å)	Pass?	Q-score
9L4B (ATR)	3.20	0.61	Yes	0.528
2YM8 (CHK1)	2.07	0.76	Yes	n/a
8P75 (CAK)	2.00	1.03	Yes	0.609
9L40 (ATR)	2.87	2.53	No	0.553
8P72 (CAK)	1.90	2.70	No	0.707
8P6Z (CAK)	2.10	4.84	No	0.705
8P6V (CAK)	1.90	6.14	No	0.738
8P7L (CAK)	2.10	6.16	No	0.708
8P74 (CAK)	2.20	6.67	No	0.628
8P6W (CAK)	1.90	7.33	No	0.762

Table 4: Summary of redocking outcomes

	<i>n</i>	Pass (rate)	95% CI
CASF-2016 subset (exploratory control)	20	8 (40%)	19.1–63.9%
CAK series (single-target)	14	1 (7.1%)	0.2–33.9%
ATR pair (9L40/9L4B)	2	1 (50%)	1.3–98.7%

40% rate was not statistically distinguishable from the literature Vina value of 58% on the full 285-complex core set [21] (one-sided binomial $p = 0.08$), though this comparison is limited by small sample size and protocol differences. The 58% reference value represents Vina performance in a randomized-ligand redocking experiment and should not be interpreted as a universal “Vina success rate,” as other CASF-2016 evaluations using different protocols report substantially higher top-pose success rates (Section S1).

The CAK series (1/14 passed, 7.1%; exact 95% CI 0.2–33.9%) showed fewer successful redockings than the exploratory CASF subset (8/20, 40%). Fisher’s exact test gave $p = 0.037$ for the prespecified one-sided comparison (motivated by the hypothesis that CAK success would be lower than the CASF control) and $p = 0.050$ for the corresponding two-sided comparison. The CASF control did not reveal an obvious gross failure of the docking implementation, whereas 13 of 14 structures in the sampled CAK series failed the prespecified criterion.

3.4 Matched flexible-sidechain analysis

Five active-site residues within 4.5 Å of native berzosertib (prespecified from the native complex) were allowed to move with Vina’s flexible-sidechain feature. Box size was held constant at 22 Å for both the flexible run and its matched rigid control. Allowing the selected active-site side chains to move partially improved pose reproduction: flexible-sidechain docking reduced RMSD from a matched rigid control of 4.77±0.06 Å (22 Å rigid only) to 3.48±0.88 Å (22 Å flexible). Sensitivity analyses using 3 residues (3.92±0.91 Å) and all residues within 5.0 Å

Table 5: Largest rank shifts between 9L40 and 9L4B scoring (matched three-seed design, $n = 40$)

Compound	Rank (9L40)	Rank (9L4B)	Shift
deriv_017	5	34	29
deriv_010	10	39	29
deriv_022	32	7	25
deriv_033	34	9	25

(3.31 ± 0.94 Å) showed similar partial improvements (Supplementary Table S5). These results indicate that local receptor flexibility can contribute to the observed failure, although the experiment does not partition receptor conformational effects from other consequences of the flexible-docking model.

3.5 Error-propagation comparison

Docking the 40-compound pilot set against the excluded 9L40 instead of validated 9L4B produced low rank concordance: Spearman $\rho = 0.21$ (bootstrap 95% CI -0.19 to 0.52), Kendall $\tau = 0.09$, mean absolute rank displacement 11.7 positions (max 29), and only 5 of 10 compounds shared between the two top-10 lists (Table 4). The five shared top-10 compounds were: camonsertib, ceralasertib, deriv_002, deriv_003, and deriv_019, the same compounds that occupied the top five ranks in the 9L4B ranking. These results show that receptor choice can alter compound prioritization within this pilot set; they do not establish a general loss of screening performance.

3.6 Implications for practice

Cognate redocking provides a structure-specific quality-control test that can identify docking failures not apparent from nominal resolution or the evaluated global map/model metrics. 9L40 and 9L4B were solved in the same study, yet only the lower-resolution one passed—an inversion that no proxy metric captured. The observation was reproduced in a separate CAK structure series, although the CAK structures derive from a single target system and therefore do not establish target-independent generality. The 13/14 observed failures in the sampled CAK series should be treated as descriptive evidence within that series rather than as a kinase-wide failure frequency.

No statistically significant association between the available global map/model metrics and redocking outcome was detected in this small survey. Because only two structures with available Q-score measurements passed, the Q-score analysis in particular was underpowered and should be interpreted as exploratory. These results support the use of per-structure cognate redocking as a

quality-control step, while systematic multi-target evaluation is required to determine its general utility.

3.7 Limitations

This is a small pilot ($n = 17$); the CAK set is a single-target convenience sample. The global Q-score analysis is underpowered (2 passing structures with reported Q-scores). Local density-fit metrics were assessed for 9L40/9L4B but a full per-residue active-site analysis across all 17 structures was not performed; such an analysis would require uniform local validation-report data that were not consistently available. No MD, MM-GBSA, or cross-kinase selectivity was run. Nothing establishes functional modulation—only predicted binding.

4 Conclusions

Cognate redocking provides a structure-specific quality-control test that can identify failures not apparent from nominal resolution or the evaluated global map/model metrics. Applied to two ATR structures from the same study, the test passed the lower-resolution one and failed the higher-resolution one, an inversion that survives box-size sensitivity analysis and is not explained by allosteric-site proximity. Within a single-target CAK structure series, 13 of 14 entries failed the same criterion. Cognate redocking is a useful structure-specific QC test that can expose docking failures not obvious from nominal resolution or the evaluated global map/model metrics, but its necessity and sufficiency for prospective enrichment require systematic evaluation. A systematic, multi-target survey is required to establish whether the proportion of failures observed here is representative of recently deposited kinase structures generally. All code, data, and structures are openly available.

Acknowledgements

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Author contributions

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Data availability

All code, seed and derivative compound tables, redocking validation results, and pilot docking results are available at <https://github.com/ethanyyy123/research-molecule-editing>. Individual replicate RMSDs, prepared PDBQT files, and full residue-deletion lists are in Supplementary Tables S1–S6. Receptor structures are publicly available from RCSB PDB (9L40, 9L4B, 2YM8) and were retrieved via the AWS Open Data mirror ([s3://pdbsnapshots](https://pdbsnapshots)).

Declarations

Conflict of interest: [Placeholder]

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